

# Alan P. Boyle

5B322 Medical Science I  
1301 Catherine St.  
Ann Arbor, MI 48109-2218  
apboyle@umich.edu  
boylelab.org

## Education

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|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2005–2009 | <b>Doctor of Philosophy</b> , Computational Biology and Bioinformatics<br>Duke University, Durham, NC                                                                                                              |
| 2001–2005 | <b>Bachelor of Science</b> , <i>summa cum laude</i> , Biochemistry and Molecular Biology<br><b>Bachelor of Science</b> , <i>summa cum laude</i> , Computer Science<br>Mississippi State University, Starkville, MS |

## Academic Appointments

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|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025–present | <b>Professor with tenure</b> , Department of Computational Medicine & Bioinformatics<br><b>Professor</b> , Department of Human Genetics<br>University of Michigan, Ann Arbor, MI                     |
| 2020–2025    | <b>Associate Professor with tenure</b> , Department of Computational Medicine & Bioinformatics<br><b>Associate Professor</b> , Department of Human Genetics<br>University of Michigan, Ann Arbor, MI |
| 2015–2020    | <b>Assistant Professor</b> , Department of Human Genetics                                                                                                                                            |
| 2014–2020    | <b>Assistant Professor</b> , Department of Computational Medicine & Bioinformatics<br>University of Michigan, Ann Arbor, MI                                                                          |
| 2010–2014    | <b>Postdoctoral Scholar</b> , Genetics<br>Stanford University, Stanford, CA; Advisor: Dr. Michael Snyder                                                                                             |
| Spring 2010  | <b>Postdoctoral Associate</b> , Computational Biology<br>Duke University, Durham, NC; Advisor: Dr. Terrence S. Furey                                                                                 |

## Institutional Affiliations

|              |                                                    |
|--------------|----------------------------------------------------|
| 2023–present | <b>Core Member</b> , Rogel Cancer Center           |
| 2021–present | <b>Affiliate</b> , Michigan Neuroscience Institute |
| 2016–present | <b>Member</b> , Center for RNA Biomedicine         |
| 2020–2023    | <b>Affiliate Member</b> , Rogel Cancer Center      |

## Other Professional Experience

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|-----------|----------------------------------------------------------------------------------|
| 2013–2014 | <b>Consultant</b> , Color Genomics<br>Personalized medicine and genomics startup |
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## Scholarships, Fellowships, and Honors

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|-----------|-----------------------------------------------------------------------------------------|
| 2022      | Valuing our Own Award, Michigan Medicine                                                |
| 2019      | Endowment for the Basic Sciences Teaching Award                                         |
| 2018      | First Place in CAGI5 Regulation Saturation Challenge                                    |
| 2017      | NSF CAREER Award                                                                        |
| 2016      | Institutional nominee for W.M. Keck Foundation Medical Science Research Program         |
| 2016      | Institutional nominee for Searle Scholar Award                                          |
| 2015–2017 | Alfred P. Sloan Foundation Fellowship in Computational & Evolutionary Molecular Biology |
| 2013–2014 | NIH Pathway to Independence Award (K99/R00) [1K99HG007356-01]                           |
| 2012      | AAAS/Science Program for Excellence in Science                                          |
| 2005–2008 | NSF Graduate Research Fellowship                                                        |
| 2005–2009 | James B. Duke Fellowship                                                                |

|             |                                                         |
|-------------|---------------------------------------------------------|
| Summer 2004 | Mayo Clinic Summer Undergraduate Research Fellow        |
| 2003        | Barry M. Goldwater Memorial Scholarship                 |
| Summer 2003 | The Institute for Genomic Research (TIGR) Summer Fellow |
| 2001        | Robert C. Byrd Honors Scholarship                       |
| 2001        | Mississippi State University Presidential Scholarship   |
| 2001        | National Merit Scholarship                              |

## Research Support

### Active: Principal Investigator and Multi-Principal Investigator

|           |                                                                                                                                                                                                                                                                                             |                            |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 2026–2031 | R01 NS145291, NIH/NINDS<br>Characterization and functional impact of somatic numtogenesis in the human cortex<br>Characterizes somatic nuclear mitochondrial DNA insertions in the human cortex and tests their effects on gene expression, chromatin structure, aging, and brain function. | (Multi-PI: Mills, Boyle)   |
| 2026–2027 | National Ataxia Foundation Research Grant<br>FGF14 intronic GAA repeats in vestibular problems beyond SCA27B<br>Examines genetic features of the FGF14 repeat that may underlie subclinical or atypical expansions.                                                                         | (Co-PI: Burmeister, Boyle) |
| 2023–2028 | UG3/UH3 NS132084, NIH/Office of the Director<br>Molecular and Computational Tools for Identifying Somatic Mosaicism in Human Tissues<br>Develops long-read molecular and computational methods to identify somatic mosaicism in normal human tissues as part of the SMaHT Consortium.       | (Multi-PI: Mills, Boyle)   |
| 2022–2026 | R01 GM144484, NIH/NIGMS<br>Mobile element derived chromatin looping variability in human populations<br>Defines how polymorphic LTR13 integrations alter three-dimensional chromatin conformation and gene regulation.                                                                      | (PI: Boyle)                |
| 2021–2027 | U01 HG011952, NIH/NHGRI<br>Predicting the impact of genomic variation on cellular states<br>Develops models and tools that use single-cell data to predict the effects of genomic variation as part of the IGVF Consortium.                                                                 | (PI: Boyle)                |

### Active: Co-Investigator and Consultant

|           |                                                                                                                                                                    |                                                       |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| 2024–2029 | R01 DE032699, NIH/NIDCR<br>Defining the Role of HPV Integration Structures in HNSCC Molecular Heterogeneity                                                        | (Multi-PI: Brenner, Mills, Spector; Co-I with effort) |
| 2023–2028 | R01 NS099280, NIH/NINDS<br>Hexanucleotide repeat translation in ALS and Frontotemporal Dementia                                                                    | (PI: Todd; Consultant)                                |
| 2022–2027 | R01 CA260677, NIH/NCI<br>The Biology of Mutant STAT6 in Follicular Lymphoma                                                                                        | (PI: Malek; Co-I with effort)                         |
| 2022–2026 | R01 NS122165, NIH/NINDS<br>Uncover the role of H3.3-G343R mutation in shaping the DNA damage response, anti-tumor immunity, and mechanisms of resistance in glioma | (PI: Castro; Co-I with effort)                        |

### Completed: Principal Investigator and Multi-Principal Investigator

|           |                                                                                                                          |                             |
|-----------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| 2022–2025 | R21 CA257864, NIH/NCI<br>High-throughput inverted reporter assay for characterization of silencers and enhancer blockers | (PI: Boyle)                 |
| 2022–2025 | Taubman Institute Innovation Program, University of Michigan<br>Repetitive elements in human health and disease          | (Co-PI: Todd, Boyle, Mills) |
| 2017–2025 | U41/U24 HG009293, NIH/NHGRI<br>RegulomeDB: A Resource for the Human Regulome                                             | (Multi-PI: Boyle, Cherry)   |

|           |                                                                                                                                                                                                                                                                 |                                  |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| 2020–2023 | R21 HG011493, NIH/NHGRI<br>New technologies for accurate capture and sequencing of repeat-associated regions                                                                                                                                                    | (Multi-PI: Boyle, Mills)         |
| 2021–2023 | Cancer Center Discovery Award, University of Michigan<br>Direct capture of complete HPV integration sites using long-read sequencing                                                                                                                            | (PI: Boyle)                      |
| 2022      | R21 HG011493-02S1, NIH/NIA<br>New technologies for accurate capture and sequencing of repeat-associated regions (supplement)                                                                                                                                    | (Multi-PI: Boyle, Mills)         |
| 2022      | NVIDIA GPU Grant, NVIDIA Corporation                                                                                                                                                                                                                            | (PI: Boyle)                      |
| 2017–2022 | DBI-1651614, NSF/BIO/DBI<br>CAREER: Conservation of cohesin-containing cis-regulatory modules in the human and mouse lineages                                                                                                                                   | (PI: Boyle)                      |
| 2019–2020 | Precision Health Investigators Award, University of Michigan<br>Short Tandem Repeats in Precision Health and Human Disease                                                                                                                                      | (Co-PI: Todd, Boyle, Mills)      |
| 2019      | NVIDIA GPU Grant, NVIDIA Corporation                                                                                                                                                                                                                            | (PI: Boyle)                      |
| 2017–2018 | Eleanor and Larry Jackier U-M/Technion and Weizmann Collaborative Research Grant<br>Michigan-Israel Partnership for Research and Education<br>Identifying novel disease-related mutations in the genomic environments around transcription factor binding sites | (Co-PI: Boyle, Mandel-Gutfreund) |
| 2015–2017 | FG-2015-65465, Alfred P. Sloan Foundation<br>Sloan Research Fellowship in Computational and Evolutionary Molecular Biology                                                                                                                                      | (PI: Boyle)                      |
| 2013–2017 | R00 HG007356 Pathway to Independence Award (K99/R00), NIH/NHGRI<br>Global Discovery and Validation of Functional Regulatory Elements                                                                                                                            | (PI: Boyle)                      |

### Completed: Co-Investigator and Consultant

|           |                                                                                                                                                                                          |                                 |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| 2018–2026 | R01 HD093570, NIH/NICHD<br>Genetic Diagnosis of Neurodevelopmental Disorders in India                                                                                                    | (PI: Bielas; Co-I with effort)  |
| 2021–2026 | K08 HL153799, NIH/NHLBI<br>Predisposition for Lung Injury in Sepsis Survival                                                                                                             | (PI: Denstaedt; Consultant)     |
| 2021–2025 | R01 HD104680, NIH/NICHD<br>Sperm Chromatin: Implications on organismal development and fertility                                                                                         | (PI: Hammoud; Co-I with effort) |
| 2023–2025 | Michigan Alzheimer's Disease Center Developmental Project, University of Michigan<br>Explore the functional impact of transposable elements in Alzheimer's disease and related dementias | (PI: Zhou; Consultant)          |
| 2017–2024 | R35 HL135824, NIH/NHLBI<br>Using Genetics to Inform Mechanism of Cardiovascular Disease                                                                                                  | (PI: Willer; Co-I with effort)  |
| 2020–2024 | W81XWH2010336, U.S. Department of Defense/Army<br>Understanding and Enhancing the Regenerative Capacity of Skeletal Muscle to Trauma by Targeting Muscle-Nerve Synergy                   | (PI: Aguilar; Co-I with effort) |
| 2016–2020 | R01 HL130705, NIH/NHLBI<br>Large-scale human genetics to understand molecular mechanisms of atrial fibrillation and related traits                                                       | (PI: Willer; Co-I with effort)  |

## Professional Service

### Institutional and Consortium Leadership

|              |                                                     |
|--------------|-----------------------------------------------------|
| 2026–present | Michigan Neuroscience Institute Executive Committee |
|--------------|-----------------------------------------------------|

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|----------------------------|---------------------------------------------------------------------------------------------|
| 2022–present               | University of Michigan Biomedical Research Council [Standing Member; Co-Chair 2026–present] |
| 2014–2018,<br>2025–present | DCMB Admissions Committee [Chair 2025–present]                                              |
| 2024–present               | Department of Human Genetics Faculty Development Committee                                  |
| 2023–present               | R01 Bootcamp Medical School Cohort Coach                                                    |
| 2023–present               | Somatic Mosaicism across Human Tissues (SMAHT) Consortium Steering Committee                |
| 2021–present               | Impact of Genomic Variation on Function (IGVF) Consortium Steering Committee                |
| 2018–present               | DCMB Laboratory Safety Liaison                                                              |
| 2015, 2016, 2025,<br>2026  | DCMB Retreat Planning Committee Chair, including the first annual retreat                   |
| 2020–2022                  | Department of Human Genetics M.S. Admissions Committee                                      |
| 2017–2024                  | DCMB Preliminary Exam Abstract Review Committee [Chair 2018–2022]                           |
| 2019–2020                  | Department of Human Genetics Ph.D. Admissions Committee                                     |
| 2017–2020                  | Department of Human Genetics Faculty Recruitment and Promotions Committee                   |
| 2016–2020                  | DCMB Seminar Series Committee [Chair]                                                       |
| 2018–2019                  | Cellular and Molecular Biology Admissions Committee                                         |
| 2017–2019                  | Endowment for the Basic Sciences Faculty IT Committee                                       |
| 2016–2019                  | DCMB Faculty Recruitment Committee                                                          |
| 2015–2017                  | Department of Human Genetics Computational Support Committee                                |
| 2008–2009                  | Duke Computational Biology and Bioinformatics Student Committee                             |

### Editorial and Program Committee Service

|            |                                                                                                          |
|------------|----------------------------------------------------------------------------------------------------------|
| 2026       | Academic Editor, PLOS ONE                                                                                |
| 2026       | Program Committee, Proceedings, ISMB/ECCB                                                                |
| 2023       | Program Committee, Genome Sequence Analysis, ISMB/ECCB                                                   |
| 2023       | Program Committee, Biomedical Informatics, ISMB/ECCB                                                     |
| 2018, 2020 | Program Committee, Comparative and Functional Genomics, ISMB/ECCB                                        |
| 2018, 2019 | Program Committee, Studies of Phenotypes and Clinical Applications, ISMB/ECCB                            |
| 2019       | Program Committee, General Computational Biology, ISMB/ECCB                                              |
| 2017       | Program Committee, Regulatory Genomics Special Interest Group Meeting (RegGenSIG), ISMB/ECCB             |
| 2015–2018  | Program Committee, Great Lakes Bioinformatics and Canadian Computational Biology Conference (GLBIO/CCBC) |
| 2015–2016  | Program Committee, Algorithms for Computational Biology (ALCOB)                                          |
| 2013–2016  | Program Committee, Gene Regulation and Transcriptomics, ISMB/ECCB                                        |
| 2012–2015  | DNA Day Essay Contest Detailed Review Judge, American Society of Human Genetics                          |

### Grant Review

|      |                                                                                                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 | NIH Study Section BDMA - Biodata Management and Analysis Study Section (ad hoc)                                                                                   |
| 2026 | NIH Study Section ZRG1 IIDB Z (70) - Stage I New Innovator Award Program (mail review)                                                                            |
| 2026 | NIH Study Section ZRG1 MGG-F (90) S - Special Topics in Genomics, Genetic Evolution, and Technology Development (ad hoc)                                          |
| 2025 | NIH Study Section BDMA - Biodata Management and Analysis Study Section (ad hoc)                                                                                   |
| 2024 | NIH Study Section ZRG1 BBOT-D (50) - PAR Panel: Biodata Management and Common Fund Data Sets (ad hoc; Co-Chair)                                                   |
| 2024 | NIH Study Section BDMA - Biodata Management and Analysis Study Section (ad hoc)                                                                                   |
| 2023 | NSF Review Panel - Molecular and Cellular Biosciences, Genetic Mechanisms (ad hoc)                                                                                |
| 2023 | NIH Study Section - Multi-Omics of Health and Disease Data Analysis and Coordination Center                                                                       |
| 2023 | NIH Study Section GVE - Genetic Variation and Evolution Study Section (ad hoc)                                                                                    |
| 2022 | NIH Study Section ZRG1 ISB-S (57) - Academic-Industrial Partnerships for Translation of Technologies for Diagnosis and Treatment                                  |
| 2022 | NASA Study Section E.11 Space Biology: Animal Studies - Omics Systems [21SBAS-OmicsSys] (ad hoc)                                                                  |
| 2020 | NIH/NIMH Study Section ZMH1 ERB-C (08) - Fine-Mapping Genome-Wide Associated Loci to Identify Proximate Causal Mechanisms of Serious Mental Illness               |
| 2019 | NIH/NIMH Study Section ZMH1 ERB-C (01) - PsychENCODE: Noncoding Functional Elements in the Human Brain and Their Role in the Development of Psychiatric Disorders |

|           |                                                                                                          |
|-----------|----------------------------------------------------------------------------------------------------------|
| 2018–2019 | University of Michigan internal review, Searle Scholars Program                                          |
| 2015      | UK Medical Research Council Methodology Research Panel (ad hoc)                                          |
| 2015      | UK Biotechnology and Biological Sciences Research Council (ad hoc)                                       |
| 2015      | Michigan Institute for Clinical and Health Research Postdoctoral Translational Scholars Program (ad hoc) |

## Manuscript Review

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|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2009–present | <i>Ad hoc</i> reviewer with <a href="#">more than 100 verified reviews</a> for journals including <i>Science</i> , <i>Nature Biotechnology</i> , <i>Nature Genetics</i> , <i>Genome Research</i> , <i>Genome Biology</i> , <i>Nature Communications</i> , <i>Bioinformatics</i> , <i>Nucleic Acids Research</i> , and <i>PLOS Computational Biology</i> |
| 2012         | Distinguished contributor as a leading reviewer for <i>Bioinformatics</i>                                                                                                                                                                                                                                                                               |

## Professional Memberships

|              |                                                     |
|--------------|-----------------------------------------------------|
| 2018–present | American Society of Human Genetics                  |
| 2013–present | International Society for Computational Biology     |
| 2012–present | American Association for the Advancement of Science |
| 2005–present | Gamma Sigma Delta Agricultural Honor Society        |

## Teaching and Mentorship

### Training Programs

|              |                                                                                    |
|--------------|------------------------------------------------------------------------------------|
| 2024–present | <b>Member</b> , Systems & Integrative Biology Training Grant (SIB)                 |
| 2021–present | <b>Member</b> , Biomedical Informatics and Data Science Training Program (BIDS-TP) |
| 2017–present | <b>Member</b> , Cellular and Molecular Biology Program                             |
| 2015–present | <b>Member</b> , Genome Science Training Program (GSTP)                             |
| 2015–present | <b>Member</b> , Michigan Predoctoral Training Program in Genetics (GTP)            |
| 2014–present | <b>Member</b> , Program in Biomedical Sciences                                     |
| 2014–present | <b>Member</b> , Bioinformatics Training Program                                    |

### Teaching (F = fall, W = winter, S = summer)

|                                              |                                                                                                 |
|----------------------------------------------|-------------------------------------------------------------------------------------------------|
| W19, W20, W21,<br>W22, W23, W24,<br>W25, W26 | Bioinformatics Concepts and Algorithms (BIOINF 529) [Course Director]                           |
| F15, F16, F17,<br>F18, F19, F20, F21         | Gene Structure and Regulation (HUMGEN 541) [3 lectures + 2 discussions / yr.]                   |
| F19, F22                                     | Research Responsibility and Ethics (PIBS 503) [1 discussion / yr.]                              |
| F21, W22, F24,<br>W25                        | Genetics Student Seminar (HUMGEN 821/822) [Mentor]                                              |
| F17, F18                                     | Experimental Genetics Systems (HUMGEN 632) [Course Director]                                    |
| F15, W16, F16,<br>W17, F17, W18,<br>F18      | Bioinformatics Journal Club (BIOINF 602/603) [Course Director F18]                              |
| S17, S18                                     | Introduction to Biocomputing Bootcamp (BIOSTAT/BIOINF/HUMGEN 606) [2 full days / yr.]           |
| F15, F16, F17                                | Introduction to Bioinformatics & Computational Biology (BIOINF 527) [2 lectures + 3 labs / yr.] |
| S15, S16, S17                                | Basic Biology for Graduate Students with Quantitative Training (BIOINF 523) [2 lectures / yr.]  |
| F03                                          | Lab TA for Isotopes Tech I (Mississippi State University, BCH 4414)                             |

### Guest Lectures / Panels

|           |                                                                                                                           |
|-----------|---------------------------------------------------------------------------------------------------------------------------|
| 2018–2019 | Lecturer, REU Site: Mathematical and Theoretical Biology Institute (MTBI), Arizona State University (NSF1757968) [2 days] |
| 2017      | Panel member, University of Michigan “New Faculty Orientation to Corporate & Foundation Relations” [ 70 attendees]        |
| 2016      | Experimental Genetics Systems (HUMGEN 632) [1 discussion]                                                                 |
| 2014      | Panel member, BIOINF 527 “Challenges in Biology, Biomedicine, Data & Analysis”                                            |
| 2010      | Co-taught Cold Spring Harbor Systems Biology Pre-meeting Workshop                                                         |
| 2009      | Duke student panelist for “How to prepare for and get into graduate school”                                               |

2008 | Taught Duke mini-course on Genome Browsers & Databases

## Mentorship

### Ph.D. Students (n=21)

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026–present | Kobe Howcroft (Ph.D. Student, Bioinformatics, University of Michigan)                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 2024–present | Ingrid Flaspohler (Ph.D. Student, Bioinformatics, University of Michigan)<br><i>NIH Biomedical Informatics and Data Science Training Program (T32)</i><br><i>Rackham Graduate Student Research Grant (pre-candidate)</i><br><i>Rackham Graduate Student Research Grant (candidate)</i>                                                                                                                                                                                                                                     |
| 2024–present | Steve Losh (Ph.D. Student, Bioinformatics, University of Michigan)                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 2024–present | Sowmya Srinivasan (Ph.D. Student, Genetics and Genomics, University of Michigan)                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2022–present | Katarina Pavlovic (Ph.D. Student, Bioinformatics, University of Michigan)<br><i>Rackham Graduate Student Research Grant (pre-candidate)</i><br><i>Rackham Graduate Student Research Grant (candidate)</i>                                                                                                                                                                                                                                                                                                                  |
| 2022–present | Rintsen Sherpa (Ph.D. Student, Bioinformatics, University of Michigan)<br><i>Rackham Graduate Student Research Grant (candidate)</i>                                                                                                                                                                                                                                                                                                                                                                                       |
| 2021–2025    | Kinsey Van Deynze (Ph.D. Student, Bioinformatics, University of Michigan)<br><i>NIH Genome Science Training Program (T32)</i><br><i>Rackham Graduate Student Research Grant (pre-candidate)</i><br><i>Rackham Graduate Student Research Grant (candidate)</i>                                                                                                                                                                                                                                                              |
| 2020–2025    | Andrea Valenzuela (Ph.D. Student, Chemical Biology, University of Michigan)<br><i>NIH Cellular Biotechnology Training Program (T32)</i>                                                                                                                                                                                                                                                                                                                                                                                    |
| 2020–2026    | Breanna McBean (Ph.D. Student, Genetics and Genomics, University of Michigan)<br><i>Joint M.S. in Bioinformatics, University of Michigan</i><br><i>NIH Genome Science Training Program (T32)</i><br><i>Rackham Graduate Student Research Grant (pre-candidate)</i><br><i>Rackham Graduate Student Research Grant (candidate)</i>                                                                                                                                                                                           |
| 2020–2025    | Camille Mumm (Ph.D. Student, Genetics and Genomics, University of Michigan)<br><i>Joint M.S. in Bioinformatics, University of Michigan</i><br><i>NIH Genome Science Training Program (T32)</i><br><i>Rackham Graduate Student Research Grant (pre-candidate)</i><br><i>Rackham Graduate Student Research Grant (candidate)</i><br><i>Rackham Predoctoral Fellowship</i>                                                                                                                                                    |
| 2018–2024    | Bradley Crone (Ph.D. Student, Bioinformatics, University of Michigan)<br><i>Rackham Graduate Student Research Grant (candidate)</i>                                                                                                                                                                                                                                                                                                                                                                                        |
| 2017–2023    | Melissa Englund (Ph.D. Student, Genetics and Genomics, University of Michigan)<br><i>NIH Human Genetics Training Program (T32)</i><br><i>Rackham Graduate Student Research Grant (candidate)</i>                                                                                                                                                                                                                                                                                                                           |
| 2018–2023    | Nanxiang (Samuel) Zhao (Ph.D. Student, Bioinformatics, University of Michigan)<br><i>Rackham Graduate Student Research Grant (pre-candidate)</i><br><i>Rackham Graduate Student Research Grant (candidate)</i>                                                                                                                                                                                                                                                                                                             |
| 2016–2018    | Haley Amemiya (Ph.D. Student, Cellular and Molecular Biology, University of Michigan)<br><i>Joint M.S. in Bioinformatics, University of Michigan</i><br><i>NIH Cellular &amp; Molecular Biology Training Program (T32)</i><br><i>NIH Cellular Biotechnology Training Program (T32) (Declined)</i><br><i>PIBS Excellence in Service Award</i><br><i>Rackham Graduate Student Research Grant (pre-candidate)</i><br><i>Rackham Graduate Student Research Grant (candidate)</i><br><i>Maas Professional Development Award</i> |

|           |                                                                                                                                                                                                                                                                                                                                                           |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           | <i>Rackham Graduate School Scholar-Activist Award</i>                                                                                                                                                                                                                                                                                                     |
| 2016–2020 | Shriya Sethuraman (Ph.D. Student, Bioinformatics, University of Michigan)                                                                                                                                                                                                                                                                                 |
| 2016–2023 | Christopher Castro (Ph.D. Student, Bioinformatics, University of Michigan)<br><i>NIH Bioinformatics Training Program (T32)</i><br><i>Rackham Merit Fellow</i><br><i>Rackham Graduate Student Research Grant (pre-candidate)</i><br><i>Rackham Graduate Student Research Grant (candidate)</i><br><i>Global Research Engagement Opportunity Fellowship</i> |
| 2017–2022 | Ningxin Ouyang (Ph.D. Student, Bioinformatics, University of Michigan)<br><i>Rackham Graduate Student Research Grant (candidate)</i>                                                                                                                                                                                                                      |
| 2016–2021 | Shengcheng Dong (Ph.D. Student, Bioinformatics, University of Michigan)<br><i>Rackham Graduate Student Research Grant (candidate)</i>                                                                                                                                                                                                                     |
| 2015–2021 | Torrin McDonald (Ph.D. Student, Genetics and Genomics, University of Michigan)<br><i>NIH Human Genetics Training Program (T32)</i><br><i>Rackham Graduate Student Research Grant (pre-candidate)</i><br><i>Rackham Graduate Student Research Grant (candidate)</i>                                                                                        |
| 2015–2017 | Greg Farnum (Ph.D. Student, Cellular and Molecular Biology, University of Michigan)                                                                                                                                                                                                                                                                       |
| 2015–2020 | Sierra Nishizaki (Ph.D. Student, Genetics and Genomics, University of Michigan)<br><i>Joint M.S. in Bioinformatics, University of Michigan</i><br><i>NIH Genome Science Training Program (T32)</i><br><i>Rackham Merit Fellow</i><br><i>Rackham Summer Award</i><br><i>Rackham Graduate Student Research Grant (candidate)</i>                            |

### M.S. Students (n=7)

|           |                                                                               |
|-----------|-------------------------------------------------------------------------------|
| 2025      | Christine Geng (M.S. Student, Bioinformatics, University of Michigan)         |
| 2023      | Hawra Aljawad (M.S. Student, Chemical Engineering, University of Michigan)    |
| 2023      | Xinyi Liu (M.S. Student, Bioinformatics, University of Michigan)              |
| 2022–2023 | Emily Pogson (M.S. Student, Genetics and Genomics, University of Michigan)    |
| 2019–2020 | Monica Holmes (M.S. Student, Bioinformatics, University of Michigan)          |
| 2017–2018 | Nanxiang (Samuel) Zhao (M.S. Student, Bioinformatics, University of Michigan) |
| 2015–2017 | Ningxin Ouyang (M.S. Student, Bioinformatics, University of Michigan)         |

### Additional Graduate Rotation Students (n=17)

|      |                                                                                               |
|------|-----------------------------------------------------------------------------------------------|
| 2024 | Karan Smith (Rotation Student, Cellular and Molecular Biology, University of Michigan)        |
| 2024 | Jun Sik Yun (Rotation Student, Genetics and Genomics, University of Michigan)                 |
| 2024 | Jeremy Chen (Rotation Student, Bioinformatics, University of Michigan)                        |
| 2023 | Rosina Carr (Rotation Student, Bioinformatics, University of Michigan)                        |
| 2023 | Connor Ward (Rotation Student, Medical Scientist Training Program, University of Michigan)    |
| 2022 | Brandt Bessell (Rotation Student, Bioinformatics, University of Michigan)                     |
| 2022 | Xiaomeng Du (Rotation Student, Bioinformatics, University of Michigan)                        |
| 2022 | Mahnoor Gondal (Rotation Student, Bioinformatics, University of Michigan)                     |
| 2022 | Xin Li (Rotation Student, Biological Chemistry, University of Michigan)                       |
| 2022 | Bohan Chen (Rotation Student, Cell and Developmental Biology, University of Michigan)         |
| 2021 | Amelia Lauth (Rotation Student, Cellular and Molecular Biology, University of Michigan)       |
| 2019 | Margarita Brovkina (Rotation Student, Cellular and Molecular Biology, University of Michigan) |
| 2018 | Steve Ho (Rotation Student, Human Genetics, University of Michigan)                           |
| 2018 | Matthew Pun (Rotation Student, Medical Scientist Training Program, University of Michigan)    |
| 2017 | Amanda Moccia (Rotation Student, Human Genetics, University of Michigan)                      |
| 2017 | Stephen Carney (Rotation Student, Human Genetics, University of Michigan)                     |
| 2016 | Tingyang Li (Rotation Student, Bioinformatics, University of Michigan)                        |

**Postdoctoral Fellows (n=4)**

|              |                                            |
|--------------|--------------------------------------------|
| 2026–present | Kinsey Van Deynze (University of Michigan) |
| 2023–2025    | Melissa Englund (University of Michigan)   |
| 2022–2025    | Torrin McDonald (University of Michigan)   |
| 2021–2022    | Shengcheng Dong (University of Michigan)   |

**Non-student Lab Volunteers (n=2)**

|           |                                                 |
|-----------|-------------------------------------------------|
| 2019–2021 | Greg Farnum (University of Michigan)            |
| 2018–2019 | Monica Holmes (Postbac, University of Michigan) |

**Undergraduate and High School Students (n=28)**

|              |                                                                                                             |
|--------------|-------------------------------------------------------------------------------------------------------------|
| 2026         | Rose Mills (High School)                                                                                    |
| 2026–present | Pria Chandarana (Undergraduate, Neuroscience, University of Michigan)                                       |
| 2026–present | Cecelia Crandall (Undergraduate, Cognitive Science, University of Michigan)                                 |
| 2026–present | Boutheine Teyeb (Undergraduate, Computer Science, University of Michigan)                                   |
| 2026         | Aanshi Agrawal (Undergraduate, Molecular, Cellular, and Developmental Biology, University of Michigan)      |
| 2025–2026    | Krrish Thakker (Undergraduate, Biochemistry, University of Michigan)                                        |
| 2023–2024    | Kateri Darr (Undergraduate, Computer Science, University of Michigan)                                       |
| 2023         | Mason Miller (Undergraduate, Computer Science, University of Michigan)                                      |
| 2022–2024    | Summer Ann (Undergraduate, Neuroscience, University of Michigan)                                            |
| 2022–2025    | Kobe Howcroft (Undergraduate, Computer Science, University of Michigan)                                     |
| 2021–2024    | Preston Parana (Undergraduate, UROP Molecular, Cellular, and Developmental Biology, University of Michigan) |
|              | <i>UROP Blue Ribbon Award</i>                                                                               |
| 2021–2022    | Julia Tweadey (Undergraduate, LSA Honors Program, Life Science Informatics, University of Michigan)         |
| 2021         | Aryn Booker (Undergraduate, UROP Molecular, Cellular, and Developmental Biology, University of Michigan)    |
|              | <i>UROP Blue Ribbon Award</i>                                                                               |
| 2020         | Marcela Alcaide Aligio (Undergraduate, SROP, Hunter College CUNY)                                           |
| 2019–2020    | David Wang (Undergraduate, UROP Computer Science, University of Michigan)                                   |
| 2019–2020    | Jack Lu (Undergraduate, UROP Computer Science, University of Michigan)                                      |
| 2019–2020    | Diana Davis (Undergraduate, Neuroscience and German, University of Michigan)                                |
| 2019         | Sheila Rasouli (Undergraduate, Neuroscience, University of Toronto)                                         |
| 2019         | Vibhasri Davuluri (High School, Girls Who Code Summer Intern)                                               |
| 2016–2019    | Cody Morterud (Undergraduate, UROP Computer Science / Honors Capstone, University of Michigan)              |
| 2016–2017    | Colten Williams (Undergraduate, UROP Computer Science, University of Michigan)                              |
| 2016–2017    | Courtney Asman (Undergraduate, Neuroscience, University of Michigan)                                        |
| 2014–2017    | Maxwell Spadafore (Undergraduate, LS&A Honors Informatics, University of Michigan)                          |
| 2013–2014    | Natalie Ng (High School, Stanford Institutes of Medicine Summer Research)                                   |
| 2013–2014    | Dana Wyman (Undergraduate, Biology, Stanford University)                                                    |
| 2013         | Justin Young (High School, Stanford Institutes of Medicine Summer Research)                                 |
| 2012         | Melanie Connick (Undergraduate, Biology, University of New Mexico)                                          |
| 2012         | Edward Dai (Undergraduate, Computer Science, Stanford University)                                           |

**Doctoral Thesis Committees (n=53)**

|              |                                                                                  |
|--------------|----------------------------------------------------------------------------------|
| 2026–present | Connor Davison (Genetics and Genomics, University of Michigan, Committee Member) |
| 2026–present | Kobe Howcroft (Bioinformatics, University of Michigan, Chair)                    |
| 2025–present | Brandt Bessell (Bioinformatics, University of Michigan, Committee Member)        |
| 2024–present | Weizhou Qian (Bioinformatics, University of Michigan, Committee Member)          |
| 2024–present | Jinhao Wang (Bioinformatics, University of Michigan, Committee Member)           |

|              |                                                                                                                                                                                                                                  |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024–present | Brandon Klein (Medicinal Chemistry, University of Michigan, Committee Member)                                                                                                                                                    |
| 2024–present | Bohan Chen (Bioinformatics, University of Michigan, Committee Member)                                                                                                                                                            |
| 2024–present | Sowmya Srinivasan (Genetics and Genomics, University of Michigan, Co-Chair)                                                                                                                                                      |
| 2024–present | Ingrid Flaspohler (Bioinformatics, University of Michigan, Chair)                                                                                                                                                                |
| 2024–present | Steve Losh (Bioinformatics, University of Michigan, Chair)                                                                                                                                                                       |
| 2024–present | Lu Lu (Bioinformatics, University of Michigan, Committee Member)                                                                                                                                                                 |
| 2023–present | Linghua Jiang (Bioinformatics, University of Michigan, Committee Member)                                                                                                                                                         |
| 2023–present | Elysia Chou (Bioinformatics, University of Michigan, Committee Member)                                                                                                                                                           |
| 2023–present | Rebecca McAvoy (Molecular, Cellular, and Developmental Biology, University of Michigan, Committee Member)                                                                                                                        |
| 2023–present | Chinmay Raut (Bioinformatics, University of Michigan, Committee Member)                                                                                                                                                          |
| 2022–present | Katarina Pavlovic (Bioinformatics, University of Michigan, Chair)                                                                                                                                                                |
| 2022–present | Rintsen Sherpa (Bioinformatics, University of Michigan, Chair)                                                                                                                                                                   |
| 2022–present | Emily Peirent (Neuroscience, University of Michigan, Committee Member)                                                                                                                                                           |
| 2025–present | Monica Holmes (Bioinformatics, University of Michigan, Committee Member)<br><i>Single-Molecule Long-read Sequencing for Resolution of Structurally Complex Genomic Loci and Accurate Phenotype Prediction</i>                    |
| 2024–2026    | Matthew Hodgman (Bioinformatics, University of Michigan, Committee Member)<br><i>Raising the Standard of Machine Learning Transparency and Utility in Cardiovascular Medicine</i>                                                |
| 2020–2026    | Breanna McBean (Genetics and Genomics, University of Michigan, Co-Chair)<br><i>Targeting Resistance and Metastasis: Advancing Local and Systemic Therapeutic Strategies for Aggressive Breast Cancers</i>                        |
| 2021–2026    | Mashiat Rabbani (Genetics and Genomics, University of Michigan, Committee Member)<br><i>A Genome-scale Program for Building the Sperm Epigenome</i>                                                                              |
| 2024–2025    | Lingrui Cai (Bioinformatics, University of Michigan, Committee Member)<br><i>Advancing Medical Image and Report Analysis with Artificial Intelligence: Applications in Inflammatory Bowel Disease and Traumatic Brain Injury</i> |
| 2021–2025    | Kinsey Van Deynze (Bioinformatics, University of Michigan, Chair)<br><i>Sequence Analysis of Tandem Repeats Utilizing 3rd Generation Sequencing Technologies</i>                                                                 |
| 2020–2025    | Andrea Valenzuela (Chemical Biology, University of Michigan, Co-Chair)<br><i>AI-Driven Workflows Integrating Strain Diversity and Host Metabolic-Genetic Variation for Tuberculosis Treatment Prioritization</i>                 |
| 2021–2025    | Zhenhao Zhang (Bioinformatics, University of Michigan, Committee Member)<br><i>Decoding Genome Regulation through Deep Learning Models</i>                                                                                       |
| 2020–2025    | Steve Ho (Genetics and Genomics, University of Michigan, Committee Member)<br><i>Applied Machine Learning for Big Data Genomics</i>                                                                                              |
| 2020–2025    | Camille Mumm (Genetics and Genomics, University of Michigan, Chair)<br><i>Exploring Repetitive Elements in Neurodegenerative Disease Using Targeted Long-Read Sequencing</i>                                                     |
| 2022–2025    | Kaiwen Deng (Bioinformatics, University of Michigan, Committee Member)<br><i>Advancing Computational Models and Algorithms for the Analysis of Single-Cell and Neural Data</i>                                                   |
| 2022–2024    | Franco Tavella (Biophysics, University of Michigan, Committee Member)<br><i>Robustness and Tunability of Biological Oscillations</i>                                                                                             |
| 2018–2024    | Bradley Crone (Bioinformatics, University of Michigan, Chair)<br><i>Computational Methods in Functional Prioritization of Polygenic Risk Score Models</i>                                                                        |
| 2021–2024    | Wenjin Gu (Bioinformatics, University of Michigan, Committee Member)<br><i>Development of Viral Integration Analysis Technologies for Virus-Associated Cancer Research</i>                                                       |
| 2018–2023    | Rucheng Diao (Bioinformatics, University of Michigan, Committee Member)<br><i>Local Chromatin Environments Shape Transcription and Adaptive Immunity in Bacteria</i>                                                             |
| 2021–2023    | Zijun Gao (Bioinformatics, University of Michigan, Committee Member)<br><i>Advance Machine Learning and Image Analysis Methods for Clinical Decision Support in Cardiovascular and Pulmonary Diseases</i>                        |
| 2018–2023    | Nanxiang (Samuel) Zhao (Bioinformatics, University of Michigan, Chair)<br><i>Decoding Regulatory Variants with Computational Methods in Non-coding Regions of the Human Genome</i>                                               |
| 2020–2023    | Ashley Melnick (Cellular and Molecular Biology, University of Michigan, Committee Member)<br><i>Cdc73 Protects Notch-Induced Leukemia Cells From DNA Damage and Mitochondrial Stress</i>                                         |

- 2016–2023 Christopher Castro (Bioinformatics, University of Michigan, Chair)  
*Investigating the Role of Noncoding De Novo Single-Nucleotide Variants in Autism Spectrum Disorder*
- 2017–2023 Melissa Englund (Genetics and Genomics, University of Michigan, Chair)  
*Identification and Characterization of Cis-Regulatory Elements in the Human Genome*
- 2018–2023 Stephen Carney (Cancer Biology, University of Michigan, Committee Member)  
*Epigenetic reprogramming in mutant IDH1 glioma influences radioresistance and neural lineage differentiation*
- 2019–2023 Benjamin Yang (Biomedical Engineering, University of Michigan, Committee Member)  
*Towards Defining Principles of Cell Fate Plasticity*
- 2018–2022 Marcus Sherman (Bioinformatics, University of Michigan, Committee Member)  
*Cultivation of enhanced bioinformatic-specific pedagogical manipulatives, interventions, and professional development*
- 2021–2022 Kuan-Han Hank Wu (Bioinformatics, University of Michigan, Committee Member)  
*Integrating Electronic Health Records with Genetic Information to Advance Precision Medicine Approaches in Cardiovascular Disease*
- 2017–2022 Amanda Moccia (Genetics and Genomics, University of Michigan, Committee Member)  
*Investigation of Developmental Disorders: Genetic Discovery and Functional Validation*
- 2017–2022 Ningxin Ouyang (Bioinformatics, University of Michigan, Chair)  
*Deciphering Transcriptional Regulatory Circuits: Transcription Factor Binding and Regulatory Variants Identification*
- 2015–2021 Torrin McDonald (Genetics and Genomics, University of Michigan, Chair)  
*Leveraging New Technologies to Explore Regulatory and Structural Elements of the Human Genome*
- 2018–2021 Heming Yao (Bioinformatics, University of Michigan, Committee Member)  
*Machine Learning and Image Processing for Clinical Outcome Prediction: Applications in Medical Data from Patients with Traumatic Brain Injury, Ulcerative Colitis, and Heart Failure*
- 2016–2021 Mohd Hafiz Bin Mohd Rothi (Molecular, Cellular, and Developmental Biology, University of Michigan, Committee Member)  
*Control of Chromatin by RNA-mediated Transcriptional Silencing*
- 2016–2021 Shengcheng Dong (Bioinformatics, University of Michigan, Chair)  
*Computational Methods to Identify Regulatory Variants in the Non-coding Regions of the Human Genome*
- 2017–2021 Steven Romanelli (Molecular & Integrative Physiology, University of Michigan, Committee Member)  
*Viral CRISPR/Cas9 Gene Transfer for Somatic Knockout in Brown Adipose Tissue*
- 2018–2021 Negar Farzaneh (Bioinformatics, University of Michigan, Committee Member)  
*Automated Decision Support System for Traumatic Injuries*
- 2016–2020 Shriya Sethuraman (Bioinformatics, University of Michigan, Co-Chair)  
*Genome-wide Identification of Non-coding Transcription by RNA Polymerase V and Its Involvement in Transcriptional Gene Silencing*
- 2015–2020 Sierra Nishizaki (Genetics and Genomics, University of Michigan, Chair)  
*Decoding the Non-coding Genome: Novel Technologies for the Characterization of Non-coding Elements and Variation*
- 2017–2020 Christopher Lee (Biostatistics, University of Michigan, Committee Member)  
*Improvements and Developments in Gene Regulation and Single-Cell Gene Expression Data Analysis*
- 2018–2019 Christine Ziegler (Biological Chemistry, University of Michigan, Committee Member)
- 2015–2018 Ari Allyn-Feuer (Bioinformatics, University of Michigan, Committee Member)  
*The Pharmacoepigenomics Informatics Pipeline and H-GREEN Hi-C Compiler: Discovering Pharmacogenomic Variants and Pathways with the Epigenome and Spatial Genome*
- 2015–2017 Raymond Cavalcante (Bioinformatics, University of Michigan, Committee Member)  
*Beyond the Transcriptome: Facilitating Interpretation of Epigenomics and Metabolomics Data*
- 2015–2017 Zhengting Zou (Bioinformatics, University of Michigan, Committee Member)  
*Model-based genomic studies of protein sequence evolution: convergence, epistasis, and amino acid acceptance rates*

**Preliminary Exam Committees (n=46)**

|      |                                                                                         |
|------|-----------------------------------------------------------------------------------------|
| 2026 | James George (Cellular and Molecular Biology, University of Michigan)                   |
| 2025 | Xinyi Deng (Bioinformatics, University of Michigan)                                     |
| 2025 | Mitchell Witt (Bioinformatics, University of Michigan)                                  |
| 2024 | Luke Gohmann (Cellular and Molecular Biology, University of Michigan)                   |
| 2024 | Benjamin Li (Bioinformatics, University of Michigan)                                    |
| 2024 | Tiffany Wan (Bioinformatics, University of Michigan)                                    |
| 2024 | Zhiyuan Yu (Bioinformatics, University of Michigan)                                     |
| 2024 | Rebecca McAvoy (Molecular, Cellular, and Developmental Biology, University of Michigan) |
| 2024 | Bonje Obua (Cellular and Molecular Biology, University of Michigan)                     |
| 2024 | Abigail Vallie (Cellular and Molecular Biology, University of Michigan)                 |
| 2023 | Jinhao Wang (Bioinformatics, University of Michigan)                                    |
| 2023 | Lishi Yin (Bioinformatics, University of Michigan)                                      |
| 2023 | Matthew Hodgman (Bioinformatics, University of Michigan)                                |
| 2023 | Ilakkiya Venkatachalam (Genetics and Genomics, University of Michigan)                  |
| 2023 | Jianhui Gong (Bioinformatics, University of Michigan)                                   |
| 2023 | Mahnoor Gondal (Bioinformatics, University of Michigan)                                 |
| 2023 | Elysia Chou (Bioinformatics, University of Michigan)                                    |
| 2022 | Sean Moran (Bioinformatics, University of Michigan)                                     |
| 2022 | Lu Lu (Bioinformatics, University of Michigan)                                          |
| 2022 | Linghua Jiang (Bioinformatics, University of Michigan)                                  |
| 2022 | Kaiwen Deng (Bioinformatics, University of Michigan)                                    |
| 2022 | Yufeng Zhang (Bioinformatics, University of Michigan)                                   |
| 2021 | Anthony Nguyen (Human Genetics, University of Michigan)                                 |
| 2021 | Hanbyul Cho (Bioinformatics, University of Michigan)                                    |
| 2021 | Charles Ryan (Cellular and Molecular Biology, University of Michigan)                   |
| 2021 | Kuan-Han Wu (Bioinformatics, University of Michigan)                                    |
| 2021 | Wenjin Gu (Bioinformatics, University of Michigan)                                      |
| 2020 | Jie Cao (Bioinformatics, University of Michigan)                                        |
| 2020 | Zijun Gao (Bioinformatics, University of Michigan)                                      |
| 2020 | Ashley Melnick (Cellular and Molecular Biology, University of Michigan)                 |
| 2019 | Benjamin Yang (Biomedical Engineering, University of Michigan)                          |
| 2019 | Maria Virgilio (Cellular and Molecular Biology, University of Michigan)                 |
| 2018 | Zhi Carrie Li (Bioinformatics, University of Michigan)                                  |
| 2018 | Kevin Hu (Bioinformatics, University of Michigan)                                       |
| 2018 | Siyu Liu (Bioinformatics, University of Michigan)                                       |
| 2018 | Alexandra Weber (Bioinformatics, University of Michigan)                                |
| 2018 | Mitch Fernandez (Bioinformatics, University of Michigan)                                |
| 2017 | Tingyang Li (Bioinformatics, University of Michigan)                                    |
| 2017 | Marcus Sherman (Bioinformatics, University of Michigan)                                 |
| 2017 | Adrienne Shami (Human Genetics, University of Michigan)                                 |
| 2017 | Trenton Frisbie (Human Genetics, University of Michigan)                                |
| 2017 | Melissa Englund (Human Genetics, University of Michigan)                                |
| 2017 | Peter Orchard (Bioinformatics, University of Michigan)                                  |
| 2017 | Li Guan (Bioinformatics, University of Michigan)                                        |
| 2016 | Shriya Sethuraman (Bioinformatics, University of Michigan)                              |
| 2016 | Jed Carlson (Bioinformatics, University of Michigan)                                    |

**Publications**

\* co-first authorship; † co-senior authorship; underline indicates Boyle Lab members

- [88] The Somatic Mosaicism across Human Tissues Network. “Integrated map of somatic mosaicism across human tissues in 25 individuals.” *bioRxiv* 2026. doi: [10.64898/2026.09.01.748636](https://doi.org/10.64898/2026.09.01.748636).
- [87] The Somatic Mosaicism across Human Tissues Network (SMaHT). “Comprehensive benchmarking of somatic mutation detection by the SMaHT Network.” *Cell* 2026, in press. PMID: [41279200](https://pubmed.ncbi.nlm.nih.gov/41279200/).

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- [84] Diehl AG, **Boyle AP**. "GPatch enables chromosome-scale, gap-free pseudoassemblies from fragmented draft genomes." *bioRxiv* 2026. PMID: 40501647.
- [83] Li S, Chou E, Wang K, **Boyle AP**, Sartor MA. "TFBSpedia: a comprehensive human and mouse transcription factor binding sites database." *bioRxiv* 2026. doi: 10.64898/2026.03.04.709638.
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- [80] Zhao N, Sherpa RN, Dong S, Howcroft KB, **Boyle AP**. "Organ-specific prioritization and annotation of non-coding regulatory variants in the human genome." *bioRxiv* 2026. doi: 10.1101/2023.09.07.556700.
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## Patents

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